

Thermodynamic State Vector Baseline Structurer & Trophic Cascade Architecture in the Web of Life Simulation Engine

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<https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 026 Technical Report

Abstract

This paper presents the theoretical framework and software architecture implemented in Sprint 026 of the *Web of Life* simulation platform. We detail the introduction of the Thermodynamic State Vector Baseline Structurer (`src/thermodynamics/state_vector.ts`), which establishes lightweight builder functions for initializing valid thermodynamic state vectors with default ambient temperatures ($T_0 = 288.15$ K) and zeroed flux records. By formalizing First Law energy conservation and Second Law entropy generation across multi-trophic interactions, our deterministic monad pipeline (`TrophicMonad`) accurately simulates solar-driven primary production, consumer assimilation efficiencies, metabolic heat dissipation, and biogeochemical cycling without violating mass-balance invariants.

1 Introduction and Systems Ecology Framing

Ecosystems are quintessential non-equilibrium thermodynamic systems. They maintain internal order by dissipating free energy received from external solar influx, subsequently releasing thermal energy into environmental sinks. Computational models of ecosystems often struggle with thermodynamic drift, wherein energy or matter is spuriously created or destroyed.

Sprint 026 addresses this by embedding physical laws directly into the core data structures and execution monads of the *Web of Life* engine. The repository (<https://github.com/pascalranoroarijaona/WebOfLife>) now enforces strict conservation contracts across all biological and abiotic components.

2 Thermodynamic Compliance & Conservation Laws

The simulation operates under two foundational thermodynamic constraints:

1. **First Law (Energy & Matter Conservation):** Total system energy E_{sys} and elemental mass (carbon C and nitrogen N) remain invariant across all transactions:

$$\frac{d}{dt} \left(\sum_{i \in S} E_i \right) = E_{solar.in} - \dot{Q}_{dissipated} = 0 \quad (1)$$

2. **Second Law (Entropy Generation):** Every metabolic process, basal respiration, and unassimilated ingestion event generates entropy through thermal dissipation:

$$\Delta S_{gen} = \frac{\dot{Q}_{dissipated}}{T_{env}} \geq 0 \quad (2)$$

where $T_{env} = 288.15$ K represents the standard baseline ambient temperature.

3 Core Architecture & State Vector Baseline

The newly structured `state_vector.ts` module provides robust builder functions to instantiate thermodynamic state vectors containing free energy pools, enthalpy metrics, and flux records.

The class hierarchy segregates entities into abiotic pools (`SolarSource`, `Atmosphere`, `SoilMatrix`) and biotic organisms (`Autotroph`, `Heterotroph`). State transitions are executed via deterministic monad pipelines (`TrophicMonad`), ensuring that carbon and energy transfers between producers and consumers satisfy stoichiometric constraints.

4 Verification & Future Work

Automated test suites assert that carbon and nitrogen pools across the soil matrix and organism biomass remain conserved within floating-point tolerances ($\epsilon = 10^{-9}$) across extended simulation horizons. Future sprints will expand spatial heterogeneity and dynamic climate feedbacks.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>